Messy Data Project - Health

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Purpose of document

Attempting to simulate what an ideal dataset would look like for addict search project.

Creating the data

```
set.seed(219)

n <- 10000
data <- tibble(
  id = 1:n,
  r = rbinom(n, 1, 0.5),
  prescribe = rbinom(n, 1, 0.2 + 0.5 * r),
  gamma = rnorm(n, mean = 5, sd = 2),
  xi = rnorm(n, mean = 0, sd = 1),
  sigma = rnorm(n, mean = 0, sd = 1),
  eps_d = rnorm(n, mean = 0, sd = 1),
  eps_a = rnorm(n, mean = 0, sd = 1),
  pi0 = 0,
  pjt = rnorm(n, mean = 50, sd = 10),
  ij = rnorm(n, mean = 5, sd = 2)
)

data <- data %>%
  mutate(
    rd_p = r + eps_d,
    ra_p = r + eps_a,
    s = rnorm(n, mean = 10, sd = 3),
    c = gamma * pjt + r^(1 / gamma) * sum(s),
```

```r
    uc = ((c^(1 - sigma)) / (1 - sigma)),
    gamma_p = gamma + ra_p * rnorm(n, mean = 0, sd = 0.5),
    profit_doc = r^2 * sum(c) + xi - (ij * r)
  )

# Quick look at the data
head(data)
```

```
# A tibble: 6 x 18
     id     r prescribe gamma     xi   sigma    eps_d   eps_a   pi0   pjt    ij
  <int> <int>     <int> <dbl>  <dbl>   <dbl>    <dbl>   <dbl> <dbl> <dbl> <dbl>
1     1     0         0  3.50 -0.438 -0.341  -0.365    1.27       0  40.1  4.42
2     2     0         0  3.64  0.635  0.0194  0.187   -0.0528     0  67.3  5.68
3     3     0         0  6.41 -1.37   0.944  -0.470    1.44       0  33.2  7.30
4     4     1         1  5.30 -0.588  0.763  -0.0165  -0.502     0  54.5  6.96
5     5     0         0  4.05 -1.74   0.259  -0.00406  0.783      0  46.8  4.47
6     6     1         1  4.66 -0.746 -0.429  -1.31     0.651      0  61.6  2.28
# i 7 more variables: rd_p <dbl>, ra_p <dbl>, s <dbl>, c <dbl>, uc <dbl>,
#   gamma_p <dbl>, profit_doc <dbl>
```

Start to optimize prescription decisions based on simulated profit and utility

```r
data <- data %>%
  mutate(
    prescribe_optimal = ifelse(profit_doc > 0 & uc > 0, 1, 0)
  )
# Summary of prescription decisions
table(data$prescribe, data$prescribe_optimal)
```

```
     0    1
  0  841 1291
  1  738 2907
```

Optimize over addict search

```
data <- data %>%
  mutate(
    addict_search = ifelse(uc < 0 & prescribe_optimal == 1, 1, 0)
  )
# Summary of addict search decisions
table(data$prescribe_optimal, data$addict_search)
```

```
       0
  0 1579
  1 4198
```